A Validated Intrinsic Rank-Two Peripheral Complement

Perron Kernel, Spectral Haar Closure, and Exact Bulk Trace Factorization

Liang Wang

July 2026

Abstract

The negative resonance of a folded-Gaussian Markov operator has recently been converted into a validated intrinsic weighted-Riesz kernel. The other peripheral mode is the Perron eigenvalue 1. Although strong positivity already makes that branch simple and isolated, a rank-two subtraction requires more: an explicit Euclidean contour, validated stored factors, a smooth continuum kernel, and simultaneous control of full and sparse matrix families. We close all four requirements at fixed noise $\sigma = 10^{-2}$.

For the Perron circle $|z-1| = 0.05$, componentwise outward Grushin certificates prove that the stored 2048, 4096, and 8192 factors are genuine spectral weighted terms, with Euclidean errors at most 2.17×10^{-10}, 4.22×10^{-10}, and 8.34×10^{-10}. A Hilbert–Galerkin and infinite-complement Schur continuation gives

$$\sup_{|z-1|=0.05} \left\| (z-\mathcal{K})^{-1} \right\|_{L^2 \to L^2} \leq 81.30575843422578.$$

The corresponding weighted term is the Perron projection itself. It has the intrinsic kernel $q_+(x, y) = \pi(y)$, where $\mathcal{K}^*\pi = \pi$ and $\int_0^1 \pi = 1$. We enclose q_+ and its derivatives in Hilbert–Schmidt norms and place it in an $L^2([0,1]^2)$ ball of radius 1.463768820944639 about the lifted archived 4096 center.

Adding the validated negative kernel q_- produces a real smooth, gauge-free rank-two kernel

$$q_{\mathrm{per}}(x, y) = \pi(y) + q_-(x, y).$$

The two contours are disjoint and each contains one algebraically simple eigenvalue. The actual stored rank-two Haar ratios now lie within 6.06×10^{-4} of the quarter–half targets. For every $n \geq 65536$, full and fixed/adaptive sparse exact-real midpoint matrices have one eigenvalue in each contour. Their union-contour Euclidean resolvent upper is 267.81252084743886, the full-to-sparse rank-two weighted-term difference is at most $9.269128034310783 \times 10^{-10}$, and the intrinsically deflated difference is at most $9.271085367195988 \times 10^{-10}$.

Finally, if $\mathcal{B} = \mathcal{K} - \mathcal{Q}_+ - \mathcal{Q}_-$, then the Perron and parity eigenvalues are replaced by zero while the remaining spectrum is unchanged. For every $m \geq 1$,

$$\mathcal{B}^m = \mathcal{K}^m - \mathcal{Q}_+ - \lambda_-^{m-1}\mathcal{Q}_-, \qquad \mathrm{tr}(\mathcal{B}^m) = \mathrm{tr}(\mathcal{K}^m) - 1 - \lambda_-^m,$$

and

$$\det(\mathrm{I} - z\mathcal{K}) = (1-z)(1-z\lambda_-)\det(\mathrm{I} - z\mathcal{B}).$$

These are structural operator factorizations, not an arithmetic trace formula. No zero-noise, zeta-zero, self-adjoint Hilbert–Pólya, or Riemann-hypothesis claim is made.

1 Introduction

The first operation in a peripheral spectral analysis should be intrinsic: remove the isolated spectral terms without choosing signs, scales, or a persistent ordering of left and right eigenvectors.

For a nonnormal operator T and a contour Γ in its resolvent set, the appropriate object is the weighted Riesz term

$$\mathcal{Q}(T;\Gamma) = \frac{1}{2\pi i}\int_\Gamma z(z-T)^{-1}\,dz = T\mathcal{P}(T;\Gamma). \tag{1}$$

For a simple eigenvalue λ, this is $\lambda r\ell^T/(\ell^T r)$, but the contour formula carries no gauge.

The analytic weighted-Riesz framework of Wang [11] showed that simple isolated smooth Nyström branches have a second-order intrinsic continuum limit. Strong positivity made the Perron instance unconditional, while the negative branch still required a validated continuum contour. That negative contour was established in Wang [6, 7]; the resulting weighted term was then constructed as a real smooth continuum kernel in Wang [10]. The parity paper also validated all three stored negative factors and supplied an all-grid full/sparse Euclidean contour from $n = 65536$.

At first sight the Perron half of the rank-two subtraction appears easier. The Markov identity gives $\mathcal{K}\mathbf{1} = \mathbf{1}$, and strong positivity gives a unique positive stationary density. Those qualitative facts are not yet a quantitative rank-two archive. Five gaps remain.

1. The exact stored Perron factors must be shown to be spectral terms, rather than accurate floating eigensolver outputs.
2. A fixed explicit circle about 1 must be transported from the stored 4096 matrix to the continuum L^2 operator.
3. The stationary density must be packaged as an intrinsic smooth Hilbert–Schmidt kernel with explicit derivative bounds.
4. The Perron and parity terms must be joined on a disjoint union contour and transferred uniformly to full and sparse families.
5. The resulting complement should be related exactly to powers, traces, determinants, and the previously studied physical two-step matrices.

This paper closes those gaps. The main conceptual simplification is that the Perron weighted term is not merely rank one: its right factor is known exactly. If π is normalized by $\langle \pi, \mathbf{1}\rangle = 1$, then

$$\mathcal{Q}_+ f = \mathbf{1}\langle \pi, f\rangle, \qquad q_+(x,y) = \pi(y). \tag{2}$$

All source derivatives vanish. This both sharpens the kernel envelope and explains why the Perron contribution changes the E and B Haar blocks but not the leading C and D tensors.

The second simplification is algebraic. Weighted terms for disjoint simple spectral components commute and annihilate one another. Consequently the bulk complement

$$\mathcal{B} = \mathcal{K} - \mathcal{Q}_+ - \mathcal{Q}_- \tag{3}$$

has exact power identities. The peripheral determinant factors are removed without an approximation and without asserting any arithmetic meaning for the residual determinant.

Three evidence levels remain separate throughout.

Analytic theorem

Strong-positive Perron structure, weighted Riesz calculus, smooth kernel identities, rank-two orthogonality, power and trace formulas, Fredholm determinant factorization, and Nyström convergence.

Outward certificate

Three exact stored Perron Grushin ledgers, two-sided factor corrections, Arb rank-two Haar intervals, Hilbert–Galerkin and Schur transfers, and all-grid normalization and cutoff gates.

Displayed center

The heat map is the exact archived 4096 rank-two factor center. It is not a pointwise interval enclosure of the continuum kernel.

2 Operator, contours, and main results

Let u_c be the unique root in $(1.5, 1.6)$ of

$$u^3 - 2u^2 + 2u - 2 = 0,$$

fix $\sigma = 1/100$, and put $m(x) = 1 - u_c x^2$. Define

$$g(x,y) = e^{-(y-m(x))^2/(2\sigma^2)} + e^{-(y+m(x))^2/(2\sigma^2)}, \qquad Z(x) = \int_0^1 g(x,y)\,dy,$$

$$k(x,y) = \frac{g(x,y)}{Z(x)}, \qquad (\mathcal{K}f)(x) = \int_0^1 k(x,y)f(y)\,dy \quad \text{on } L^2([0,1]). \tag{4}$$

The kernel is real, strictly positive, and C^∞ on the compact square. Moreover

$$\mathcal{K}\mathbf{1} = \mathbf{1}. \tag{5}$$

The two contours are

$$\Gamma_+ = \{z \in \mathbb{C} : |z-1| = 0.05\}, \tag{6}$$
$$\Gamma_- = \{z \in \mathbb{C} : |z-c_-| = 0.05\}, \qquad c_- = -0.9865481927458079. \tag{7}$$

They are disjoint. Write $\Gamma_{\text{per}} = \Gamma_+ \cup \Gamma_-$ with the positive orientation on each component, and define

$$\mathcal{Q}_\pm = \frac{1}{2\pi i}\int_{\Gamma_\pm} z(z-\mathcal{K})^{-1}\,dz, \qquad \mathcal{Q}_{\text{per}} = \mathcal{Q}_+ + \mathcal{Q}_-. \tag{8}$$

For $n \geq 2$, let $h = 1/n$, $x_i = y_i = (i+\frac{1}{2})h$, and let P_n° be the continuum-normalized midpoint matrix

$$(P_n^\circ)_{ij} = hk(x_i, y_j).$$

Let P_n be the exact-real discretely row-normalized full matrix and $P_n^{(L)}$ its support-truncated and renormalized analogue. The fixed family uses $L = 8$; the adaptive family uses the RH-39 schedule once growth is needed.

Theorem 2.1 (Validated intrinsic Perron kernel)**.** *The circle Γ_+ lies in the resolvent set of $\mathcal{K}$, contains exactly one eigenvalue counted algebraically, and*

$$\sup_{z\in\Gamma_+} \left\|(z-\mathcal{K})^{-1}\right\|_2 \leq 81.30575843422578. \tag{9}$$

The enclosed eigenvalue is exactly 1 and algebraically simple. There is a unique real smooth strictly positive π satisfying

$$\mathcal{K}^*\pi = \pi, \qquad \int_0^1 \pi(y)\,dy = 1,$$

and

$$\mathcal{Q}_+ f = \mathbf{1}\langle \pi, f\rangle, \qquad q_+(x,y) = \pi(y). \tag{10}$$

In particular, $\mathcal{Q}_+$ has rank one and is both the Perron Riesz projection and its weighted term.

The kernel satisfies

$$1 \leq \|q_+\|_{L^2([0,1]^2)} \leq 4.065287921711291, \tag{11}$$

$$\|(q_+)_y\|_{L^2} \leq 1555.261017746445, \qquad \|(q_+)_{yy}\|_{L^2} \leq 192884.0796819851, \tag{12}$$

while every derivative containing x vanishes. At $n = 65536$, its midpoint sample differs from its cell-average matrix by at most

$$2.510507555832183 \times 10^{-6}. \tag{13}$$

It lies in an operator-norm ball of radius 1.035040859379391 *and an L^2-kernel ball of radius* 1.463768820944639 *about the lifted archived* 4096 *Perron factor.*

Theorem 2.2 (Validated intrinsic rank-two peripheral kernel)**.** *The union contour Γ_{per} contains exactly two eigenvalues,* 1 *and the real simple negative resonance λ_- of RH-43. The intrinsic peripheral weighted term has rank two and smooth real kernel*

$$q_{\text{per}}(x, y) = \pi(y) + q_-(x, y). \tag{14}$$

It is gauge-free and obeys

$$\mathcal{Q}_+\mathcal{Q}_- = \mathcal{Q}_-\mathcal{Q}_+ = 0. \tag{15}$$

The kernel is enclosed in an L^2 ball of radius 3.890944960739168 *about the lifted archived* 4096 *rank-two center, and*

$$\|q_{\text{per}}\|_{L^2} \le 9.919454564031339. \tag{16}$$

The stored Perron and parity factors at 2048, 4096, *and* 8192 *are all genuine spectral weighted terms. The resulting actual rank-two Haar ratios satisfy*

$$\begin{aligned} E &: [0.2497518320109983, 0.2498816430591350], \\ C &: [0.5000266705203903, 0.5000297811731884], \\ B &: [0.5000390841288479, 0.5000413558665742], \\ D &: [0.2494528393779779, 0.2506059251560737]. \end{aligned} \tag{17}$$

Theorem 2.3 (Uniform rank-two full, sparse, and bulk families)**.** *For every integer $n \ge 65536$, the exact-real full matrix P_n, the fixed eight-sigma sparse matrix, and the adaptive sparse matrix have exactly one eigenvalue in each of Γ_+ and Γ_-. On the union contour, their Euclidean resolvents are uniformly bounded by*

$$267.81252084743886. \tag{18}$$

For the full and either sparse family,

$$\left\| \mathcal{Q}_{\text{per}}(P_n) - \mathcal{Q}_{\text{per}}(P_n^{(L_n)}) \right\|_2 \le 9.269128034310783 \times 10^{-10}, \tag{19}$$

and for the intrinsically deflated matrices

$$B_n = P_n - \mathcal{Q}_+(P_n) - \mathcal{Q}_-(P_n), \tag{20}$$

the full-to-sparse difference is at most

$$9.271085367195988 \times 10^{-10}. \tag{21}$$

The full rank-two weighted term converges to the midpoint sample of q_{per} at order n^{-2}. Adaptive support preserves the rate $O(n^{-2}(\log n)^{-1/4})$, and the same rate holds for the adaptive bulk operator relative to the full bulk family.

Theorem 2.4 (Exact bulk powers, traces, and determinant)**.** *Let*

$$\mathcal{B} = \mathcal{K} - \mathcal{Q}_+ - \mathcal{Q}_-. \tag{22}$$

Then

$$\mathcal{B}\mathbf{1} = 0, \qquad \langle \pi, \mathcal{B} f \rangle = 0, \tag{23}$$

the eigenvalues 1 *and* λ_- *are replaced by zero, and every other spectral value is unchanged away from zero. For every integer* $m \geq 1$,

$$\mathcal{B}^m = \mathcal{K}^m - \mathcal{Q}_+ - \lambda_-^{m-1} \mathcal{Q}_-. \tag{24}$$

The smooth Gaussian kernel is trace class. Consequently

$$\operatorname{tr}(\mathcal{B}^m) = \operatorname{tr}(\mathcal{K}^m) - 1 - \lambda_-^m \tag{25}$$

and the Fredholm determinants satisfy

$$\det(\mathrm{I} - z\mathcal{K}) = (1 - z)(1 - z\lambda_-) \det(\mathrm{I} - z\mathcal{B}). \tag{26}$$

The same identities hold exactly for every finite matrix family for which the two contours are valid.

3 The Perron weighted term is the stationary kernel

The qualitative Perron statement is analytic and independent of the computer-assisted contour.

Proposition 3.1 (Strong-positive Markov structure)**.** *The eigenvalue* 1 *of* $\mathcal{K}$ *is algebraically simple. There exists a unique strictly positive* $\pi \in C^\infty([0,1])$ *with* $\mathcal{K}^*\pi = \pi$ *and* $\langle \pi, \mathbf{1} \rangle = 1$*. The Riesz projection at* 1 *is*

$$\mathcal{P}_+ f = \mathbf{1} \langle \pi, f \rangle. \tag{27}$$

Since the eigenvalue is 1*,* $\mathcal{Q}_+ = \mathcal{K}\mathcal{P}_+ = \mathcal{P}_+$*.*

Proof. The normalized kernel k is strictly positive and smooth on the compact square, so $\mathcal{K}$ is compact and strongly positive. The Markov identity (5) fixes its spectral radius at 1. Krein–Rutman theory and strong positivity make the eigenvalue algebraically simple and give a strictly positive dual eigenfunction [2, 4]. Normalize it by $\langle \pi, \mathbf{1} \rangle = 1$. The rank-one formula for a simple Riesz projection gives (27). Multiplication by $\mathcal{K}$ does not change that projection. □

Proposition 3.2 (Perron derivative and midpoint envelope)**.** *Let* K_y *and* K_{yy} *denote the Hilbert–Schmidt operators obtained by differentiating* $k(x, y)$ *in the target variable. Then*

$$\|\pi'\|_2 \leq \|K_y\|_{\mathrm{HS}} \|\pi\|_2 \,, \qquad \|\pi''\|_2 \leq \|K_{yy}\|_{\mathrm{HS}} \|\pi\|_2 \,. \tag{28}$$

If $Q^\circ_{+,n}$ *is the midpoint sample and* $Q^{\mathrm{G}}_{+,n}$ *the cell-average matrix of* q_+*, then*

$$\left\| Q^\circ_{+,n} - Q^{\mathrm{G}}_{+,n} \right\|_2 \leq \frac{\|\pi''\|_2}{\sqrt{320}\, n^2}. \tag{29}$$

Proof. The dual fixed-point equation is

$$\pi(y) = \int_0^1 k(x, y) \pi(x)\, dx.$$

Differentiate under the integral. The Hilbert–Schmidt operator inequality gives (28). Since $q_+(x, y) = \pi(y)$, source derivatives and all mixed derivatives vanish. The one-dimensional midpoint-average Peano kernel has L^2 norm $h^{3/2}/\sqrt{320}$ on each cell; summing the cell errors yields (29). □

On Γ_+, the Riesz projection norm is bounded by

$$\|\mathcal{P}_+\|_2 \leq 0.05 \sup_{z \in \Gamma_+} \left\| (z - \mathcal{K})^{-1} \right\|_2 .$$

Because $\|\mathbf{1}\|_2 = 1$ and $\langle \pi, \mathbf{1} \rangle = 1$,

$$1 \leq \|\pi\|_2 = \|\mathcal{P}_+\|_2 \,.$$

Combining (9) with the RH-42 derivative envelope gives (11)–(13).

4 Exact stored Perron factors

The archived matrices at $n = 2048, 4096, 8192$ contain a floating Perron eigenvalue approximation, a nearly constant right mode, and a positive left mode. Small residuals alone do not prove that the stored rank-one factor is the weighted Riesz term of the exact stored matrix. We use the two-sided correction theorem of Wang [10].

For a stored matrix A, approximate eigenvalue λ_0, and left–right factor

$$P_0 = \frac{r_0 \ell_0^T}{\ell_0^T r_0},$$

define

$$\widetilde{A} = \lambda_0 P_0 + (\mathrm{I} - P_0)A(\mathrm{I} - P_0). \tag{30}$$

Then P_0 reduces $\widetilde{A}$ exactly, and

$$\widetilde{A} - A = -(A - \lambda_0 \mathrm{I})P_0 - P_0(A - \lambda_0 \mathrm{I}) + P_0(A - \lambda_0 \mathrm{I})P_0. \tag{31}$$

The correction norm is controlled by the two residuals and the enclosed Gram factor. If $M \left\| \widetilde{A} - A \right\| < 1$ on the contour, the resolvent identity transfers $\lambda_0 P_0$ to the actual weighted Riesz term of A.

The inverse certificate treats every stored binary64 number as an exact real input. Sparse LU provides an approximate bordered inverse; componentwise outward residual arithmetic bounds its exact inverse in both the one- and infinity-norms, whose geometric mean gives a Euclidean upper [3, 9]. The radius 0.1 is too wide for this particular Grushin center: at $n = 2048$ its transport product is 1.366. The fixed radius 0.05 closes all three levels with products near 0.683.

Table 1: Exact-stored Perron Grushin and weighted-factor certificates.

n	residual upper	contour resolvent	correction product	weighted-term error
2048	$4.3058\,10^{-11}$	65.1669349545	$6.314\,10^{-11}$	$2.1601\,10^{-10}$
4096	$1.1923\,10^{-10}$	65.2291154903	$1.232\,10^{-10}$	$4.2167\,10^{-10}$
8192	$3.3310\,10^{-10}$	65.2559386375	$2.434\,10^{-10}$	$8.3379\,10^{-10}$

The centers differ from 1 by at most 2.5×10^{-15}. At the base level the exact target circle $|z - 1| = 0.05$ is obtained by a center-shift Neumann product

$$2.896754635731181 \times 10^{-14}. \tag{32}$$

The two circles enclose the same weighted term; only the resolvent upper is transported.

5 From the stored Perron circle to the continuum

The RH-42 exact-stored-to-midpoint bridge is independent of which isolated mode is being followed. Its defect is

$$\left\| P_{4096}^{\text{stored}} - P_{4096}^{\circ} \right\|_2 \leq 1.055416712190961 \times 10^{-5}. \tag{33}$$

The first Neumann transfer gives the exact-midpoint Perron-circle resolvent 65.27405269296997. The midpoint-to-cell-average Galerkin defect is $8.115839277660032 \times 10^{-4}$ and yields the 4096 Hilbert-Galerkin resolvent 68.92540148121991.

Let V_n be the cell-average space and split $V_{2n} = V_n \oplus W_n$ by the orthogonal Haar transform. Relative to this decomposition, write

$$A_{2n} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}.$$

The derivative envelope gives

$$\|C_n\| \le \frac{K_x}{\pi n}, \quad \|B_n\| \le \frac{K_y}{\pi n}, \quad \|D_n\| \le \frac{K_{xy}}{\pi^2 n^2}.$$

Since $\min_{z\in\Gamma_+} |z| = 0.95$, the detail resolvent is controlled by $1/(0.95 - \|D_n\|)$. The Schur self-energy

$$\Sigma_n(z) = B_n(z - D_n)^{-1}C_n$$

then transfers the count whenever $M_n \sup_{\Gamma_+} \|\Sigma_n(z)\| < 1$.

Table 2: Validated Perron Hilbert–Galerkin and complement Schur chain.

step	Schur/Neumann product	resulting resolvent upper
stored $\to$ exact midpoint	$6.8844\,10^{-4}$	65.2740526930
midpoint $\to V_{4096}$	$5.2976\,10^{-2}$	68.9254014812
4096 $\to$ 8192	$1.1228\,10^{-1}$	77.8066710318
8192 $\to$ 16384	$3.1675\,10^{-2}$	80.3990839395
16384 $\to$ 32768	$8.1817\,10^{-3}$	81.0793334963
32768 $\to$ 65536	$2.0627\,10^{-3}$	81.2562912048
full infinite complement	$5.1679\,10^{-4}$	81.3057584342

For the final step, decompose $L^2 = V_{65536} \oplus V_{65536}^{\perp}$. The same derivative bounds apply to the full infinite complement. Its self-energy product is only $5.167900126933194 \times 10^{-4}$, proving (9) and the count-one statement. Since $\mathcal{K}\mathbf{1} = \mathbf{1}$, the enclosed eigenvalue must be 1; strong positivity identifies it as algebraically simple.

6 Weighted transport and the Perron construction ball

The Schur resolvent formula can be integrated around Γ_+. If the detail block has no spectrum inside the circle, the weighted fine term differs from $\mathrm{diag}(\mathcal{Q}(A_n), 0)$ only through the Schur self-energy and the three coupling blocks. The Frobenius sum of those four block bounds gives an operator upper [10].

Starting from the exact stored 4096 Perron factor gives the ledger in table 3. All numbers are Euclidean operator uppers.

Table 3: Weighted Perron transport from the stored center to the continuum.

gate	weighted-term difference upper
stored factor $\to$ actual stored term	$4.2166963\,10^{-10}$
stored $\to$ exact midpoint	0.0023592032
midpoint $\to V_{4096}$	0.1916958114
4096 $\to$ 8192	0.5250689957
8192 $\to$ 16384	0.1885770690
16384 $\to$ 32768	0.0756069632
32768 $\to$ 65536	0.0347628684
finite rank $\to$ continuum complement	0.0169699485
total	1.0350408594

The difference of two rank-one operators has rank at most two, hence its Hilbert–Schmidt norm is at most $\sqrt{2}$ times its operator norm. This gives the Perron L^2 construction radius in theorem 2.1. Adding the RH-43 parity construction balls gives

$$\left\|\mathcal{Q}_{\mathrm{per}} - Q^{\mathrm{center}}_{\mathrm{per},4096}\right\|_2 \le 2.751313566962289, \tag{34}$$

$$\left\|q_{\mathrm{per}} - q^{\mathrm{center}}_{\mathrm{per},4096}\right\|_{L^2} \le 3.890944960739168. \tag{35}$$

7 The actual spectral rank-two Haar law

For a smooth kernel q, midpoint sampling and the orthogonal Haar transform give four blocks. With the orientation used in the archive,

$$h^{-2}E_h \longrightarrow \frac{q_{xx}+q_{yy}}{32}, \tag{36}$$

$$h^{-1}C_h \longrightarrow \frac{q_x}{4}, \tag{37}$$

$$h^{-1}B_h \longrightarrow \frac{q_y}{4}, \tag{38}$$

$$h^{-2}D_h \longrightarrow \frac{q_{xy}}{16} \tag{39}$$

in Hilbert–Schmidt sampling; see Wang [8, 10]. For the Perron kernel,

$$(q_+)_x = (q_+)_{xx} = (q_+)_{xy} = 0. \tag{40}$$

Thus its leading C and D tensors vanish exactly. In the combined kernel, the leading C and D tensors are precisely those of the parity kernel, whereas E and B include the stationary density.

The RH-40 Arb ledger enclosed the exact algebraic rank-two factors but did not prove that both factors were spectral. RH-43 closed the parity factor errors. The present Perron errors complete the spectral correction. If e_n is the sum of the Perron and parity factor errors at level n, then

$$\begin{aligned} e_{2048} &\le 5.810907673369557 \times 10^{-10}, \\ e_{4096} &\le 1.141956473291731 \times 10^{-9}, \\ e_{8192} &\le 2.260221050983747 \times 10^{-9}. \end{aligned} \tag{41}$$

Orthogonality of the Haar transform bounds the correction of E by $\sqrt{2}(e_n + e_{2n})$ and the other three blocks by $\sqrt{2}e_{2n}$.

Table 4: Validated intervals for the actual spectral rank-two Haar ratios.

block	actual ratio interval	maximum target deviation
E	$[0.2497518320, 0.2498816431]$	$2.482\,10^{-4}$
C	$[0.5000266705, 0.5000297812]$	$2.979\,10^{-5}$
B	$[0.5000390841, 0.5000413559]$	$4.136\,10^{-5}$
D	$[0.2494528394, 0.2506059252]$	$6.060\,10^{-4}$

The smallest block is D, so its interval widens most under the spectral correction. It nevertheless remains wholly within 10^{-3} of the target $1/4$. The quarter–half mechanism is therefore both analytic and spectral for the complete rank-two peripheral term.

8 The intrinsic rank-two kernel

Let $\mathcal{P}_\pm$ be the Riesz projections. Since the contours are disjoint, functional calculus gives

$$\mathcal{P}_+\mathcal{P}_- = \mathcal{P}_-\mathcal{P}_+ = 0, \qquad \mathcal{K}\mathcal{P}_\pm = \mathcal{P}_\pm\mathcal{K}. \tag{42}$$

Moreover

$$\mathcal{Q}_+ = \mathcal{P}_+, \qquad \mathcal{Q}_- = \lambda_- \mathcal{P}_-.$$

This proves (15). Both summands have nonzero one-dimensional ranges, so their sum has rank two. The kernel is exactly (14), independent of all eigenvector gauges.

Adding the specialized Perron envelope to the RH-43 parity envelope gives the explicit bounds in table 5.

Table 5: Hilbert–Schmidt envelope for the intrinsic rank-two kernel.

quantity	rigorous upper
$\|q_{\mathrm{per}}\|_{L^2}$	9.919454564031339
$\|(q_{\mathrm{per}})_x\|_{L^2}$	3780.968408944756
$\|(q_{\mathrm{per}})_y\|_{L^2}$	3715.926624442619
$\|(q_{\mathrm{per}})_{xx}\|_{L^2}$	$1.107398829423627\, 10^6$
$\|(q_{\mathrm{per}})_{xy}\|_{L^2}$	$1.544489270236306\, 10^6$
$\|(q_{\mathrm{per}})_{yy}\|_{L^2}$	$4.608506732586612\, 10^5$
$\|(q_{\mathrm{per}})_{xxyy}\|_{L^2}$	$5.610208521245976\, 10^{10}$

At $n = 65536$, the midpoint-to-cell-average defect is bounded by

$$2.041176267936486 \times 10^{-5}. \tag{43}$$

The displayed center in figure 1 is not a pointwise interval enclosure. The validated object is the Hilbert–Schmidt ball and derivative envelope.

9 Uniform full, sparse, and rank-two-deflated families

At $n = 65536$, the continuum-to-midpoint defect for the Markov kernel is 0.005112929739977326. Applying it to the Perron contour gives product 0.4157106303297654 and exact-midpoint resolvent 139.15323922479328. Discrete row normalization then gives the full matrix resolvent

$$139.27331158261282. \tag{44}$$

The eight-sigma cutoff defect is

$$\varepsilon_{65536} \le 1.957332885203985 \times 10^{-13},$$

and the sparse Perron resolvent is 139.27331158640953.

For two operators on one circle, the resolvent identity gives

$$\|\mathcal{Q}(A;\Gamma) - \mathcal{Q}(B;\Gamma)\| \le r R_\Gamma M_A M_B \|A - B\|. \tag{45}$$

Hence the Perron full-to-sparse weighted difference is

$$1.993240948303590 \times 10^{-10}. \tag{46}$$

Adding the RH-43 parity bound yields (19); adding the matrix cutoff once yields (21).

The parity contour remains the limiting condition. Its uniform sparse resolvent upper is 267.81252084743886, larger than the Perron upper. Therefore the disjoint union contour inherits exactly the RH-43 threshold $n \ge 65536$ and the same maximum resolvent constant.

The analytic Nyström theorem of Wang [11], now with both continuum premises closed, gives unconditionally

$$\left\| \mathcal{Q}_{\mathrm{per}}(P_n) - Q^\circ_{\mathrm{per},n} \right\|_2 = O(n^{-2}). \tag{47}$$

Combining this with the adaptive RH-39 cutoff gives

$$\left\| \mathcal{Q}_{\mathrm{per}}(P_n^{(L_n)}) - Q^\circ_{\mathrm{per},n} \right\|_2 = O\Big(n^{-2}(\log n)^{-1/4}\Big). \tag{48}$$

Remark 9.1 (Fixed versus adaptive support). The fixed eight-sigma rank-two difference is uniformly below 9.28×10^{-10}, sufficient for the present contour and deflation theorems. The underlying fixed-width row defect has a nonzero continuum floor. Adaptive growth is required when convergence to the full continuum kernel is claimed.

10 Exact bulk algebra, traces, and determinants

The validated rank-two term now permits an exact intrinsic bulk operator. Its algebra is stronger than a perturbative subtraction.

Proposition 10.1 (Peripheral annihilation and powers). *Let $\mathcal{B}$ be defined by (22). Then (23) holds and, for every $m \geq 1$, (24) holds.*

Proof. The Markov and stationary identities give

$$\mathcal{K}\mathbf{1} = \mathbf{1}, \qquad \mathcal{Q}_+\mathbf{1} = \mathbf{1}, \qquad \mathcal{Q}_-\mathbf{1} = 0,$$

and their dual analogues give the left constraint. Functional calculus gives

$$\begin{aligned} \mathcal{K}\mathcal{Q}_+ &= \mathcal{Q}_+\mathcal{K} = \mathcal{Q}_+, & \mathcal{Q}_+^2 &= \mathcal{Q}_+, \\ \mathcal{K}\mathcal{Q}_- &= \mathcal{Q}_-\mathcal{K} = \lambda_-\mathcal{Q}_-, & \mathcal{Q}_-^2 &= \lambda_-\mathcal{Q}_-, \end{aligned}$$

together with (15). Thus $\mathcal{B}$ vanishes on the two peripheral ranges and equals $\mathcal{K}$ on the invariant complementary spectral subspace. Raising this block decomposition to the mth power proves (24). □

The case $m = 2$ is particularly relevant to the archived physical matrices:

$$\mathcal{B}^2 = \mathcal{K}^2 - \mathcal{Q}_+ - \lambda_-\mathcal{Q}_-. \tag{49}$$

Thus the earlier Perron/parity-extracted physical two-step construction has an exact intrinsic continuum target. From (47)–(48) and

$$\left\| B_n^2 - \mathcal{B}^2 \right\| \leq (\|B_n\| + \|\mathcal{B}\|) \|B_n - \mathcal{B}\|,$$

the full and adaptive physical two-step families inherit the same rates, provided the matrix and continuum operators are compared in the standard cell basis.

Proposition 10.2 (Trace and determinant factorization). *The identities (25) and (26) hold.*

Proof. The Gaussian kernel is C^∞ on a compact one-dimensional square, so the associated integral operator is trace class; finite-rank subtraction preserves trace class [1, 5]. Taking traces in (24) uses

$$\operatorname{tr}(\mathcal{Q}_+) = 1, \qquad \operatorname{tr}(\lambda_-^{m-1}\mathcal{Q}_-) = \lambda_-^m,$$

and gives (25). Relative to the invariant direct sum

$$L^2 = \operatorname{Ran}\mathcal{P}_+ \dotplus \operatorname{Ran}\mathcal{P}_- \dotplus \mathcal{H}_{\text{bulk}},$$

$\mathcal{K}$ has diagonal blocks 1, λ_-, and the bulk restriction, whereas $\mathcal{B}$ has blocks 0, 0, and the same bulk restriction. The Fredholm determinant is multiplicative over this direct sum, proving (26). □

Remark 10.3 (What the factorization does not say). Equation (26) removes two validated noisy peripheral factors. It does not identify the remaining determinant with an arithmetic zeta function, a prime-power trace, or the spectrum of a self-adjoint operator. It is the correct structural input for such later questions, not their conclusion.

11 Certificate summary and figure

The complete certificate closes the following gates.

1. exact stored Perron Euclidean Grushin counts at 2048, 4096, and 8192;
2. two-sided Perron factor errors below 8.34×10^{-10};
3. a continuum Perron contour with resolvent upper 81.306;
4. an intrinsic stationary kernel and a rank-two continuum kernel;
5. actual spectral rank-two quarter–half intervals;
6. an all-grid two-contour theorem from $n = 65536$;
7. rank-two weighted and deflated cutoff bounds below 9.28×10^{-10};
8. exact bulk power, trace, and Fredholm determinant factorizations.

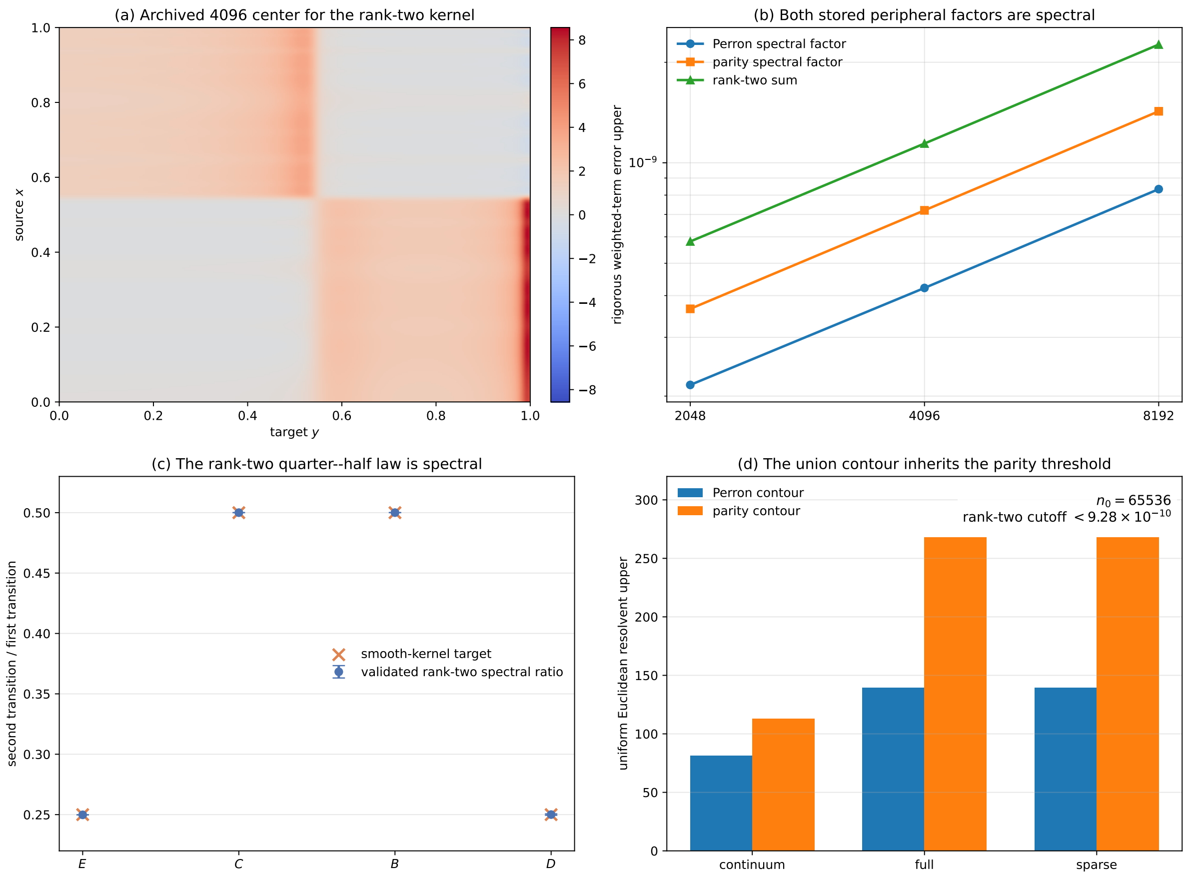

Figure 1: Validated intrinsic rank-two peripheral closure. (a) The exact archived 4096 Perron-plus-parity factor center, displayed as a piecewise kernel; it is not a pointwise interval enclosure. (b) Actual weighted-term errors for the stored Perron and parity factors and their rank-two sum. (c) Corrected Frobenius intervals for the actual spectral rank-two Haar ratios. (d) Perron and parity contour resolvent constants; the disjoint union inherits the parity threshold and maximum.

12 What is closed and what remains

The present result changes the status of the peripheral route in five ways.

1. **Both continuum peripheral modes are intrinsic kernels.** The Perron kernel is the stationary density in the target variable; the negative kernel is the RH-43 weighted parity term.
2. **The union contour is quantitative.** Each component has count one and a dimension-uniform Euclidean resolvent for full and sparse families.
3. **All six stored factors are spectral.** Perron and parity factors at three levels now have exact-stored Grushin and two-sided correction certificates.
4. **The rank-two quarter–half law is actual.** Its intervals concern true matrix weighted terms, not only archived low-rank algebra.
5. **The bulk operator has exact trace algebra.** Peripheral subtraction commutes with every power and removes the two elementary Fredholm determinant factors exactly.

Several boundaries remain essential.

1. The noise width is fixed at $\sigma = 1/100$.
2. The exact continuum kernels are enclosed in Hilbert–Schmidt norms; no pointwise interval heat map is claimed.
3. The all-dimension theorem concerns exact-real Gaussian formulas. The binary64 theorem concerns the six archived factors.
4. Fixed eight-sigma support is spectrally stable but is not claimed to converge to the full kernel in row norm.
5. The determinant factorization is structural. No arithmetic trace formula, prime-power identity, zeta-zero identification, $T \log T$ counting law, self-adjoint Hilbert–Pólya operator, or Riemann-hypothesis conclusion follows.

The natural next gate is now sharply defined: prove trace-norm convergence of the intrinsic bulk two-step family and construct its validated Fredholm determinant, without yet imposing an arithmetic interpretation. Unlike the earlier route, that step no longer carries dimension-dependent peripheral eigenvector gauges.

13 Archive and reproducibility

The archive contains two principal ledgers:

1. the multilevel exact-stored Perron Euclidean Grushin certificate; and
2. the complete Perron contour, factor, rank-two Haar, continuum kernel, full/sparse family, cutoff, and bulk composition certificate.

Every cross-paper input, local source, result, figure, and publication artifact is recorded by SHA-256.

The complete replay is:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_multilevel_perron_grushin.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_rank_two_certificate.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 \
  /root/math/.venv/bin/python experiments/make_figures.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  -m pytest -q -p no:cacheprovider
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf validated-rank-two-peripheral-complement.pdf
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_archive.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/verify_archive.py
```

The 8192 bordered inverse is the longest step, taking roughly six minutes on the current server. The composition, Arb Haar ledger, tests, figures, and archive verification take seconds.

14 Conclusion

The peripheral subtraction is now complete at fixed positive noise. The Perron term is the stationary kernel $\pi(y)$; the negative term is the validated parity kernel. Their disjoint Riesz contours produce a smooth gauge-free rank-two object, and both the continuum and all sufficiently fine full/sparse matrix families carry exactly one eigenvalue on each branch.

This completion also resolves the last ambiguity in the archived rank-two Haar ledger. Every stored Perron and parity factor is now a genuine spectral weighted term, so the corrected quarter–half intervals concern the actual matrices. The union contour does not worsen the RH-43 threshold: the parity branch remains the limiting condition.

Most importantly, the residual bulk is no longer defined by an eigensolver convention. It is the intrinsic operator $\mathcal{B} = \mathcal{K} - \mathcal{Q}_+ - \mathcal{Q}_-$, with exact power, trace, and Fredholm determinant factorizations. This does not supply an arithmetic spectrum, but it gives the next paper a clean operator: a validated rank-two-free bulk whose two-step trace and determinant can be studied without peripheral gauge or normalization artifacts.

References

[1] Israel C. Gohberg and Mark G. Krein. *Introduction to the Theory of Linear Nonselfadjoint Operators*, volume 18 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1969.

[2] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, Berlin, 2 edition, 1995. doi: 10.1007/978-3-642-66282-9.

[3] Siegfried M. Rump. Verification methods: Rigorous results using floating-point arithmetic. *Acta Numerica*, 19:287–449, 2010. doi: 10.1017/S096249291000005X.

[4] Helmut H. Schaefer. *Banach Lattices and Positive Operators*. Springer, Berlin, 1974. doi: 10.1007/978-3-642-65970-6.

[5] Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2 edition, 2005. doi: 10.1090/surv/120.

[6] Liang Wang. Validated isolation of a negative continuum resonance for a folded-gaussian markov operator: A Grushin–Rouché certificate and an exact dyadic Galerkin lift. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-41-validated-parity-continuum-contour.

[7] Liang Wang. Uniform euclidean isolation of a negative parity resonance: A hilbert–galerkin lift from the continuum operator to fixed-width and adaptive sparse Nystr"om families. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-42-uniform-euclidean-parity-contour.

[8] Liang Wang. Dyadic Haar block decay for smooth markov kernels. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-38-dyadic-haar-block-decay.

[9] Liang Wang. Outward-rounded primal–dual residual enclosures at a quadratic band-merging map. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-27-outward-rounded-primal-dual-residuals.

[10] Liang Wang. A validated intrinsic kernel for a negative parity resonance: Weighted-Riesz factor correction, complement Schur closure, and adaptive spectral deflation. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-43-validated-weighted-riesz-parity-kernel.

[11] Liang Wang. Gauge-free weighted Riesz terms for Nystr"om peripheral extraction: A second-order continuum bridge and exact stored Haar ledgers. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-40-weighted-riesz-projector-bridge.